

Autumn Mid Semester Examination, 2025

Course Code: CS531

Course Title: Advanced Algo and Data Structures

Full Marks : 30

Time : 1.5 Hours

Answer as many questions as you like but maximum you get is 40.

Figures in the right hand indicate marks for the individual question.

1. Explain whether the following statements are true or false. [6 × 3]

∴ i) $2^{2n} \in \Theta(2^{4n})$.

∴ ii) If $\lim_{n \rightarrow \infty} (\log f(n) - 2 \log g(n)) = 1$ and $f(n) \in \Theta(g(n))$.

iii) The time complexity for inserting an element into a sorted array of n elements is $\Theta(n)$.

iv) In the recursive closest pair algorithm for n points, total number of calls to **AcrossPair()** is $\frac{n}{2}$. (Recall that **AcrossPair()** is called on subarrays of length 2 or more to find the closest pair that are on different sides of the dividing line.)

v) Time complexity of the following code is $\Theta(n \log n)$.

```
Recurse (int n ) {  
    if (n==0) return n;  
    else return 1 + Recurse(n/3) + Recurse(n/9);  
}
```

∴ vi) If $f(n) = 1 + a + a^2 + a^3 + \dots + a^{2n}$ and $a < 0$, the $f(n) \in \Theta(a^{2n})$.

2. Consider a different version of the mergesort algorithm. Instead of dividing the array in two, it divides the array in three equal sized subarrays; then recursively sort them; finally merge the three sorted subarrays. Write down the recurrence to compute the time complexity $T(n)$ of the algorithm with explanation. Solve the recurrence to find $T(n)$. Assume $n = 3^k$. [7]

3. Show the working of the recursive algorithm for finding the 5th largest element in the following array. [7]

Show the contents of the array in each recursive call..

3, 4, 10, 1, 13, 7, 8, 72, 82, 19, 2, 14, 6, 11, 5, 9, 12, 15, -2

4. a) Let $A[n]$ be an **unimodal** array containing only 1s and 0s, with all 1s appearing before all 0s. For example, the following array is **unimodal**,
In this problem, design an $O(\log n)$ time algorithms to find the index of the right most 1. For the above example, the output will be 5. [5]
- b) Now consider a 2-D **unimodal** array of size $[n \times n]$ where every row is **unimodal**, as in the following figure.

1	1	0	0	0	0
1	0	0	0	0	0
1	1	1	1	1	0
0	0	0	0	0	0
1	1	1	0	0	0
1	1	1	1	0	0

- i) Devise an $O(n \log n)$ algorithm to find the index of the rightmost 1 in a 2-D **unimodal** array. [3]
- ii) Design a faster algorithm (possibly linear) than the above. Prove that nobody cannot design a better one. [5+3]
5. In the **Closest Pair** problem, the point set P is passed as P_X (sorted by x -coordinates) and P_Y (sorted by y -coordinates) to the recursive function. The point set P is then divided into two equal sized subsets P_L and P_R by a vertical line. The points in P_L also has to be partitioned to P_{LX} (sorted by x -coordinates) and P_{LY} (sorted by y -coordinates). Similarly P_R has to be divided into P_{RX} and P_{RY} .
- i) Write an algorithm to convert P_X and P_Y into P_{LX} , P_{LY} , P_{RX} and P_{RY} . Your algorithm should run in $O(n)$ time. Explain the time complexity of your algorithm. [5+3]
- ii) Validate your algorithm with an example. [5]

TEZPUR UNIVERSITY

Autumn Semester Sessional Test-I, 2025

Design and Analysis of Algorithms/ Algorithms & Complexity (CS 542/701).

Full Marks : 10

Time : 1 Hour

You may answer all the questions but maximum you can get is 10.

1. Prove or disprove whether the following statements are true or not. [4 × 2]

- (a) Multiplying two matrices $A[1 \times n]$ and $B[n \times 1]$ takes $\Theta(n^2)$ time.
- (b) If $f(n) \in O(g(n))$, then $\log f(n) = \Theta(\log g(n))$.
- (c) Given two sorted arrays A_L and A_R , each containing n elements, finding whether there is an element present in both A and B will take $\Theta(n)$ time.
- (d) $2^{n+4} \in O(2^n)$.

2. Consider k sorted arrays $\{A_0, \dots, A_{k-1}\}$. each array A_i contains 2^i elements. Design an algorithm to merge these arrays into a single sorted array A . Analyze the time complexity of your algorithm in terms of k using $\Theta(\)$ notation. [3+5]

3. Consider a town A of n inhabitants. Design a method that will find a person x of A who is not among the 10 richest, neither he is among the 5 poorest of the town A . You can request a person about the value of his/her property and compare the value with another persons wealth. How many persons will you request to get the person x ? Give a lower bound on the number of requests. [2+3]

4. Answer the following questions considering the following function.

```
1 unsigned int funct ( unsigned int n )
2 {
3     int i, j, z = 0;
4     for (i = n; i >= 1; i = i/2)
5         for (j = 1; j <= n; j=j + n/100) {
6             z = z + 1;
7         }
8     return z;
9 }
10
```

a) Express the value returned by $funct(n)$ using big-theta Θ notation for an input size n . Prove the bound. [2]

b) Analyze the time complexity of the function (using Θ notation) for n . [3]

TEZPUR UNIVERSITY

Autumn Semester End Examination, 2025

Course Code: CSMT531/501

Course Title: Adv. Algo and Data Structures

Full Marks: 50

Time: 2 hours

You may answer all the questions but maximum you can get is 50.

1. Determine the time complexity of the following code using Θ -notation. [5]

```

for (int i = 1; i ≤ n; i++)
    for (int j = n; j ≥ 1; j = j/2)
        for (int k = 1; k ≤ n; k++)
            print "Hello World";

```

Handwritten analysis: $O(n)$ for the first loop, $O(\log n)$ for the second loop, and $O(n)$ for the third loop. The total complexity is $O(\log n) * O(n) * O(n) = O(n^2 \log n)$.

2. Explain whether the following statements are True / False. Correct answers without explanation will get 0. [3×5]

- Any n node tree ($n > 2$) has atleast 2 leaf nodes.
 - If $T(n) = 3T(\lceil \frac{n-3}{5} \rceil) + O(n)$, then $T(n) \in \Theta(n^2)$.
 - The largest weighted edge of a graph G will not be in the minimum spanning tree of G .
 - Dijkstra's algorithm can find the longest path in a graph (with positive weights) from a source vertex if we multiply every edge weight with -1 .
 - An optimal solution to a 0/1 knapsack problem will always contain the object with the greatest benefit/weight ratio.
3. Assume there are n men with heights $m_1, \dots, m_n$ and there are n women with heights $w_1, \dots, w_n$. The task is to pair each man with a woman in such a way that the total height difference between the pairs is minimized. Each man or women is matched with exactly one partner. That means, the task is to minimize $\sum_{i=1}^n |m_i - s_j|$ where (i, j) are partners.

Consider the following two greedy algorithms:

- Find the man and woman with the minimum height difference. Pair them up. Repeat until every man and woman is matched.

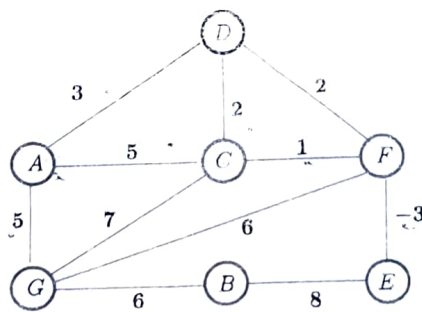
- (b) Sort the men and women by height. Match the shortest man with the shortest woman, and repeat.

Which one of the algorithms above is correct? Disprove the incorrect one with a counterexample. For the correct algorithm, provide a proof of its correctness [2+3+5]

4. Suppose we have a bag of capacity 10 kg, which we want to pack with items of values $\{10, 40, 30, 50\}$ and weights 5, 4, 6 and 3 respectively. Our goal is to pack the bag with as many items as possible so that the total weight of the items does not exceed the capacity of the bag and the total value is maximized. Solve the problem using dynamic programming (**Show the table only**). Will your algorithm work if:

- one of the item has negative weight? [1]
- one of the item has negative weight and negative profit? [7+1+2]

5. a) What is the cost of a minimum spanning tree of the graph below? [1]
 b) How many minimum spanning trees does this graph have? List them all. [3]
 c) Run Prim's algorithm on this graph. For each of the iteration, show: The SCORE of each vertex, the HEAP, the vertex extracted from the HEAP. [8]
 d) Which data structure will you use to store the MST? Explain. [2]



6. Given a message M with the symbols $\{a, b, c, d, e\}$. Huffman algorithm encodes the symbols as $a \equiv 000, b \equiv 001, c \equiv 11, d \equiv 01, e \equiv 10$. Let the frequencies of the symbol c and b are 30 and 15 respectively. Determine the frequency of the other symbols such that justifies the codes all the symbols. [5]